

Robust General_Sepsis_V11: Cohort-Aware Validation, 3D Topology, and SHAP Explainability

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Objective:
Remove optimistic validation artifacts and re-evaluate hybrid GNN architecture under robust cohort-aware splits.

Methods:
Fold-internal MAD feature selection, dataset-aware CV, threshold calibration from OOF, model-by-model comparisons, SHAP explainability, and 3D graph topology rendering.

Key Metrics:
Hybrid CV AUROC: 0.848
Hybrid CV Accuracy: 0.878
Hybrid External AUROC: 0.991
Hybrid External Accuracy: 0.796

Conclusion:
Comparisons now use robust validation and interpretability-ready artifacts for defensible claims.

